ASTANA IT UNIVERSITY

«Approved»
by the Dean
__________ Syzdykova Z.A.
«___» of __________ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	**Computer Graphics Fundamentals**
Degree cycle (level)/ major	Bachelor, IT
Year, term	3,1
Number of credits	5
Language of delivery:	English
Prerequisites	Fundamentals of programming C++, Linear Algebra Algorithms and Data Structure
Postrequisites	No Postrequisites
Lecturer(s)	Turar Olzhas, Olzhas.turar@astanait.edu.kz Astana IT University, Expo, C1 block, 2nd floor, office C1.1.336
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	This course allows students to acquire basic fundamental knowledge on how computer graphics work on low level of graphics libraries and devices. They will acquire practical skills on programming basic graphical applications using OpenGL. These can be easily scaled to high level usage of GUI based engines and graphical applications.
2. Course Goal(s)	The main purpose of this course is to teach students computer graphics techniques and libraries, including theoretical mathematical and algorithmic aspects of computer graphics, and their practical implementation using modern OpenGL graphics library.
3. Course Objectives:	• Define computer graphics and its importance in various industries. • Introduce graphics software tools and their roles in the creation and manipulation of images. • 2D Graphics • Learn about basic 2D drawing algorithms and techniques. • Understand 3D coordinate systems and transformations. • Learn about shading models, lighting, and texturing. • Introduce graphics programming languages and libraries (e.g.. OpenGL or DirectX). • Study color theory and color models used in computer graphics. • Explore interactive graphics concepts, including user interfaces and event-driven programming. • Develop interactive graphics applications and games. • Understand the role of computer graphics in multimedia presentations and virtual reality experiences. • Apply the knowledge and skills gained throughout the course to complete a substantial graphics project.

4.	Skills & Competences	• Theoretical knowledge of basic fundamentals of computer graphics • Theoretical knowledge of modern computer graphics methods and Vulkan • Theoretical knowledge of GPU architecture and usage .... • Practical skills of implementation of GPU accelerated graphics using OpenGL • Practical skills on usage of basic mathematical methods of lighting, texturing, animation, etc.
5.	Course Learning Outcomes:	Upon successful completion of the Computer Graphics Fundamentals course, students will be able to: • Explain the fundamental concepts of computer graphics and its applications across various industries. • Identify Graphics Hardware and Software Components: • Recognize and explain the key components of graphics hardware and software, including GPUs, monitors, and graphics APIs. • Produce 2D graphics using raster graphics techniques, understanding concepts like pixels, colors, and basic drawing algorithms. • Describe the principles of 3D graphics, including 3D modeling, transformations, and rendering. • Implement and analyze fundamental graphics algorithms such as line drawing, clipping, and polygon filling. • Utilize rasterization and rendering methods for both 2D and 3D graphics, including shading, lighting, and texturing. • Develop graphics applications using programming languages and libraries commonly used in computer graphics (e.g., OpenGL, DirectX). • Apply matrices and vectors to represent and manipulate graphical data effectively. • Demonstrate an understanding of color theory and image processing techniques to enhance and manipulate images.
6.	Methods of Assessment	• Individual Homeworks, each have to be protected, with code explanation • Term quiz, 30% of term grade, quiz for about 20-25 questions: 1 answer, multiple choice, text answer • Final project, implemented in team of 3 students. Require hard independent work
7.	Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic Literature: 1. OpenGL Super Bible 3rd (third) Edition by Wright, Richard S, Lipchak 2. Ogldev.com 3. Learnopengl.com

	Supplementary literature: 1. Vulkan.com 2. Vulkan-tutorial.com
1. Resources	Ogldev.com Learnopengl.com
2. **Course policy**	**Course and University policies include:** **Attendance:** Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean's office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the

	being late to the class is not welcome and can have restriction activities by the course instructor. **Attestation I and II:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. **Late submissions:** Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **Final exam* / Final project:** At the completion of this course each each team has to show working computer graphics application according to chosen topic. **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. **Online lessons** can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors.

	Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.

* *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Introduction. • Computer Graphics History. • OpenGL 1.0, 1.5 Practice on OpenGL 1.5 elements	[1], [3]	2	3	1	9
2	Modern OpenGL: • Buffer • Shader • Matrix Transformation Practice on OpenGL 3.3	[2]	2	3	1	9
3	Lighting Principles • Phong Lighting • Parametric Geometry Practice on Phong Lighting	[2]	2	3	1	9

4	Textures • Texturing • Introduction to 3D Modelling • OBJ Model Format • Interactivity Inordinary shading	[2]	2	3	1	9
5	Framebuffers Objects: • GLFW • Rendering in Texture Midterm Quizz	[2]	2	3	1	9
6	Ray Casting • 3D Textures Perlin noise	[1],[2]	2	3	1	9
7	Geometry Shaders • Billboarding Tessellation shader	[1]	2	3	1	9
8	Skeletal Animation Basics Biped Character Animation	[1],	2	3	1	9
9	Some Advanced Topics on Computer Graphics • Inverse Kinematics • Global Illumination Approaches PBR	[1],	2	3	1	9
10	Modern Graphics Approaches • Ray Tracing Vulkan	[1]	2	3	1	9
	Total hours:	**150**	**20**	**30**	**10**	**90**

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	**2**	**3**	**4**	**5**
1	OpenGL 1.5 Examples	10	Books, Lectures	Midterm quiz
2	Buffers and Shaders	10	Books, internet resources	Midterm quiz
3	Matrix Transformations, Phong Lighting	10	Books, internet resources	Midterm quiz
4	Textures	10	Books, internet resources	Midterm quiz
5	Framebuffer Objects	10	Books, internet resources	Midterm quiz
6	Ray Casting	10	Books, internet resources	Endterm quiz
7	Geometry Shader	10	Books, internet resources	Endterm quiz
8	Skeletal Animations	10	Books, internet resources	Endterm quiz
9	Skeletal Animations	10	Books, internet resources	Endterm quiz
10	Modern Graphics Approaches	9	Books, internet	Endterm quiz

			resources	

4. Student Performance Evaluation System for the Course

Period	Assignments			Number of points	Total
1st attestation	1 Assignment (week 3), 2 Assignment (week 4), 3 Assignment (week 5)	100 100 100	**70**	**30**	100
	Mid term quiz		**30**		
2nd attestation	1 Assignment (week 6), 2 Assignment (week 7), 3 Assignment (week 8), 4 Assignment (week 9)	100 100 100 100	**70**	**30**	100
	End term quiz		**30**		
Final exam	Final Project			40	100
Total	**0,3*1stAtt+0,3*2ndAtt+0,4*Final**				**100**

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1.33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment

80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	**Job title**	**Date**	**Sign**
Turar Olzhas	Associate professor	01.08.2024	

Director **Sergaziyev M.Zh.**